Roll No. ....

TPE–013

B. TECH. (ECE) (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

SATELLITE COMMUNICATIONS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) State and explain the Kepler's law for planetary motion. Discuss the difference between geosynchronous and geostationary orbit. (CO1)

   (b) What do you understand by look angle ? Explain the limits of visibility with reference to a LEO satellite and earth station. (CO1)

(c) A satellite is orbiting in geosynchronous orbit of radius 41500 km. Find the velocity and time of the orbit. What will be the change of velocity if the radius reduces to 36000 km ? If $g_o = 398600.5\ km^2/s^2$. (CO1)

2. (a) Explain the Altitude and Orbit Control (AOC) Subsystem. (CO2)

(b) Explain the equipment reliability with the help of bath tub curve, and what do you understand by space qualification. (CO2)

(c) A satellite stationed at a distance 380000 km from the surface of the earth radiates a power of 3 watts, in the direction of an earth station. Assuming the satellite antenna gain to be 20 dB. If the operating frequency is 10 GHz and gain of receiving antenna is 50 dB, calculate the power received. (CO2)

3. (a) Derive the Link Equation used for calculating the power received by radio link. (CO3)

(b) What are the various propagation impairments for satellite signal ? Explain the effect of rain on satellite link reliability. (CO3)

(c) A radio link uses a pair of 2 m dish antenna with an efficiency of 60% each, as transmitting and receiving antenna. Other specifications of the link are; Transmitted power is 1 dBW, Carrier frequency is 4 GHz, Distance of receiver from the transmitter is 150 m. Calculate : (CO3)

(i) Free Space Loss

(ii) Power gain of each Antenna

(iii) Received power in dBW

4. (a) What is the difference between multiplexing and multiple access techniques ? Explain pre-assigned and demand assigned multiple access. (CO4)

   (b) Explain different multiple access techniques in detail. (CO4)

   (c) If a satellite has transponder having 50 MHz BW and 1023 barker code is used for CDMA by various earth stations, the chip rat is 32 MCPS. Determine the number of accesses for an SNR of 12 dB. Also find the total occupancy of transponder if BPSK is used. (CO4)

5. (a) What is principle of trilateration for navigation ? Explain the NavlC satellite system. (CO5 & CO6)

   (b) What do you understand by the DTH System ? Explain the working of DTH. (CO5 & CO6)

   (c) What is the difference between regional and global navigation systems ? Explain the working of GPS. (CO5 & CO6)

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Roll No. ....

TPE–004

B. TECH. (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

DIGITAL SYSTEM USING VERILOG

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the following with the help of a suitable example : (CO1)

(i) 'timescale

(ii) $monitor

(iii) 'define

(iv) $finish

(v) $time

P. T. O.

(b) Discuss the following Verilog data types giving two examples of each : (CO1)

(i) Parameters

(ii) Memories

(iii) Strings

(iv) Time

(v) Arrays

(c) Discuss the two basic types of digital methodologies. (CO1)

2. (a) Write the Verilog description of 4-input multiplexer using : (CO2 & CO3)

(i) Gate level modelling

(ii) Data flow modelling

(iii) Behavioral modelling

(b) Explain the following operators with the help of suitable examples : (CO2 & CO3)

(i) Concatenation

(ii) Replication

(iii) Conditional

(iv) Reduction

(v) Shift

(c) Describe Function and Task in Verilog using an example code for each. Point out the difference between functions and taks. (CO2 & CO3)

3. (a) Explain the difference between : (CO2 & CO3)

(i) Initial and always statement

(ii) Sequential and parallel blocks

(b) Design a clock with time period = 40 ns and a duty cycle of 25% by using the always and initial statement. The value of clock at time = 9 should be initialized to 0. (CO2 & CO3)

(c) Write the Verilog description of JK flip-flop and verify its working using suitable test bench. (CO2 & CO3)

4. (a) What is UDP in Verilog ? How many types of UDP are there ? Explain the generic structure of UDP. (CO4)

(b) Define a negative edge triggered T flip-flop with clear as a UDP. Signal clear is active low signal. Use shorthand notation wherever possible. (CO4)

(c) Write the switch level Verilog description for nor gate. Test the described nor gate using test bench. (CO4)

5. (a) Explain the basic structure of a testbench. Why testbench module don't have any port list ? (CO5 & CO6)

(b) Write a test bench for the following : (CO5 & CO6)

(i) T-flip flop

(ii) Serial in Serial Out shift register

(c) Write a short note on Object Oriented Programming. (CO5 & CO6)

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TEC–701

B. TECH. (ECE) (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

PRINCIPLES OF MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss interpersonal and decisional roles of a manager. What are various goals or objectives of a management ? (CO1)

   (b) Explain the following qualities of a manager : (CO2)
   education, maturity, self confidence, human relation attitude, training.

   (c) What is management ? Elaborate the statement "Management is both an art and a science." Also prove that management is a continuous process. (CO1)

2. (a) Discuss *four* methods to track or measure customer satisfaction. (CO2)

   (b) What are *six* major steps in planning ? Briefly describe each of them. (CO2)

   (c) Explain all *four* factors in SWOT analysis in detail. (CO2)

P. T. O.

3. (a) Discuss *four* steps of organizing in detail with special reference to identification, grouping, assignment of duties and delegation of authority. (CO3)

   (b) Discuss the importance of HR management. (CO3)

   (c) Discuss various principles of directing in detail. (CO3)

4. (a) Define HR management and suggest means to overcome the challenges. (CO4)

   (b) What is controlling process ? Discuss *four* steps of controlling process. (CO4)

   (c) Discuss any *three* limitations in the process of controlling. (CO4)

5. (a) Suggest and explain any *three* methods to track or measure customer satisfaction. (CO5 & CO6)

   (b) Define Recruitment and discuss all the major steps in recruitment process in detail. (CO5 & CO6)

   (c) Define training. Discuss the importance of training. Differentiate among needs, wants and demands. (CO5 & CO6)

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TEC–731

B. TECH. (ECE) (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

DISASTER MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define vulnerability and discuss the vulnerability profile of India. (CO2)

(b) Explain the importance of disaster management plan in an event of epidemic like Dengue. (CO4)

(c) Explain the importance of team building in disaster management. (CO5)

2. (a) Explain the different mitigation processes with their examples. (CO2)

(b) Bring to light major problems in Relief distribution. (CO6)

(c) Describe various tools and technologies which could be used for disaster management. (CO5)

3. (a) 'The relationship between disaster and development depends on the development choices made by the individual, community and the nation'. Discuss. (CO1)

   (b) Describe the role of Social organizations in case of natural calamities. (CO5)

   (c) Write a note on deforestation as an event for disaster. (CO4)

4. (a) Write a detailed note on cyclone shelter. (CO5)

   (b) Examine the role of sustainable development in disaster management. (CO3)

   (c) Discuss the role of disaster management authorities in disaster management. (CO5)

5. (a) Examine the role of corporate social responsibility as an emerging avenue in managing disasters. (CO5)

   (b) Write a case study on any one disaster in Uttarakhand. (CO4)

   (c) Explain the role and functions of a Disaster Manager. (CO6)

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TEC–759

B. TECH. (ECE) (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

INTERNET OF THINGS AND ITS APPLICATIONS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the functional blocks of IoT ? Explain with the help of a proper diagram. (CO1)

   (b) Explain the constraints and challenges of the wireless sensor network. (CO1)

   (c) Explain the IoT Layers in detail. (CO1)

2. (a) What is M2M ? What is the M2M value chain ? How can we connect M2M with IoT ? (CO2)

   (b) Explain the outline structure of IoT Architecture. Take the example of Agriculture in support of IoT architecture. (CO2)

P. T. O.

(c) Explain the main design principles and needed capabilities for IoT. (CO2)

3. (a) Explain the Operational view and Information View in the context of IoT. (CO3)

   (b) Explain IoT Reference Model. (CO3)

   (c) What is the meaning of State of the Art in the context of UoT ? (CO3)

4. (a) How can IoT help in improving Health and Lifestyle ? (CO4)

   (b) What is the role of IoT in Energy ? (CO4)

   (c) What is the role of IoT in Industry ? (CO4)

5. (a) A Secure system is needed for the Industrial IoT purpose. What are the important things that need to consider while designing a security protocol for such a system ? (CO5, CO5 & CO6)

   (b) How are Privacy and Security important issues while designing and IoT system ? (CO5, CO5 & CO6)

   (c) Explain Privacy and Trust in IoT Data in the context of smart cities. (CO5, CO5 & CO6)

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TPE–007

B. TECH. (ECE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023

SEMICONDUCTOR DEVICES AND MODELING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain in detail about formation of the energy band. (CO1)

(b) Explain the variation of drift velocity and mobility with an electric field for semiconductors. (CO1)

(c) Show the dependence of channel length modulation parameter (λ) on drain current. (CO1)

2. (a) Explain the capacitance-voltage characteristics of two-terminal MOS structure. (CO2)

(b) Determine the voltage gain, current gain, input impedance and output impedance for RC-coupled common-source amplifier. (CO2)

P. T. O.

(c) Write short notes on the following : (CO2)

(i) Body effect in MOS Transistor

(ii) Drain-Induced Barrier Lowering (DIBL)

(iii) Gate Induced Drain Lowering (GIDL)

3. (a) Describe the Fin-FET structures and explain its working and advantage in detail. (CO3)

(b) Discuss the hot carrier effect and impact ionization in a MOSFET. (CO3)

(c) Explain SOI structures in detail. Mention the advantages of SOI structures. (CO3)

4. (a) Write a SPICE code of enhancement mode MOSFET circuit for DC analysis. (CO3 & CO4)

(b) A step graded Ge diode, having $N_D = 400\ N_A$, acceptor impurities are 2 impurity atoms for every 10^6 atoms at room temperature. Find the barrier potential. (CO3 & CO4)

Assume $n_i = 2.5 \times 10^{13}/\text{cm}^3$ and

Total number of Ge atoms = $4.4 \times 10^{22}/\text{cm}^3$

(c) Differentiate between partially depleted and fully depleted SOI MOSFETs. (CO3 & CO4)

5. (a) Describe the working principle of heterojunction bipolar transistors. (CO5 & CO6)

(b) Explain HEMTs in detail and differentiate between HBTs and HEMTs. (CO5 & CO6)

(c) Explain MOSFET structure with proper band diagram. (CO5 & CO6)

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TPE–002

B. TECH (ECE) (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

OPTICAL FIBER COMMUNICATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) The refractive indices of core and cladding materials of a step index fiber are 1.48 and 1.45 respectively.

Calculate : (CO1)

(i) numerical aperture

(ii) acceptance angle

(iii) the critical angle

(iv) fractional refractive indices change.

(b) Define the relative refractive refractive index difference for an optical fiber and show how it may be related to the numerical aperture. (CO1)

(c) A graded index fiber has a core with a parabolic refractive index profile which has a diameter of 50 μm. The fiber has a numerical aperture of 0.2. Estimate the total number of guided modes propagating in the fiber when it is operating at a wavelength of 1 μm. (CO1)

2. (a) Briefly describe the scattering losses occur in optical fiber. (CO2)

(b) Explain Mode field diameter and spot size. (CO2)

(c) When a long single mode fiber is operating at a wavelength of 1.3 mm, the attenuation is 0.5 dB/km. The core diameter of the fiber is 6 μm and the laser source bandwidth is 700 MHz. Calculate the threshold optical powers for stimulated Brillouin and Raman scattering. (CO2)

3. (a) Explain the concept of population inversion for Ruby (crystal) and He-Ne (gas) laser. (CO3)

(b) Describe heterojunction, how it performs better than normal p-n junction. (CO3)

(c) Describe with the aid of suitable diagrams the mechanism giving the emission of light from an LED. (CO3)

4. (a) When 3×10^{11} photons each with a wavelength of 0.85 um are incident on a photo diode, on average 1.2×10^{11} electrons are collected at the terminals of the device. Determine the quantum efficiency and the responsivity of the photodiode at 0.85 μm. (CO4)

(b) Define photocarriers and photocurrent. What are the advantages of photodiodes ? (CO4)

(c) What is absorption ? Explain direct and indirect absorption. (CO4)

5\. (a) What is a preamplifier ? Give the classification of preamplifier. (CO5, CO6)

(b) A 4-port multimode fiber FBT (fused bioconical tapper) coupler has 70 μW optical power launched into port one. The measured output powers at ports two, three and four are 0.004, 26 and 27.5 μW respectively. Determine : coupling ratio, excess loss, insertion loss between the input and output ports. (CO5, CO6)

(c) Explain the concept of wavelength division multiplexing in optical fiber systems. (CO5, CO6)